

Table 1: Comparison of behavioral simulation methods. EFR is the only approach that combines LLM heterogeneity with transparency while maintaining behavioral consistency.

Method	Uses LLM heterogeneity?	Trans- parent?	Inference- time?	MVR (wait)↓	Spearman ρ ↑	Out-of-design inference?
Pure-DCE (empirical frontier)	×	✓	✓	0.000	1.000	×
Uniform Shrinkage ($\lambda\bar{p} + (1 - \lambda)P_{static}$)	×	✓	✓	0.000	1.000	×
EFR (this work) ($\lambda\hat{p}_{LLM} + (1 - \lambda)P_{static}$)	✓	✓	✓	0.042	0.952	✓
Unconstrained LLM	✓	×	✓	0.625	−0.024	✓

Note: MVR (monotonicity violation rate) and Spearman’s ρ are evaluated against P_{static} on held-in DCE design points (Qwen2.5-72B, $\lambda = 0.25$). Out-of-design inference indicates whether the method can generalize to scenarios beyond the original DCE attribute space. EFR is the only method that simultaneously achieves behavioral consistency, transparency, and out-of-design capability.